CSE251 – Electronic Devices and Circuits

Rectifier

UTSAB SAHA, Lecturer
Department of Computer Science and Engineering (CSE)
BRAC University

Lecture 8: Outline

- Half-wave Rectifier
- Full-wave Rectifier
- Rectifier with a Filter Capacitor

Rectifier

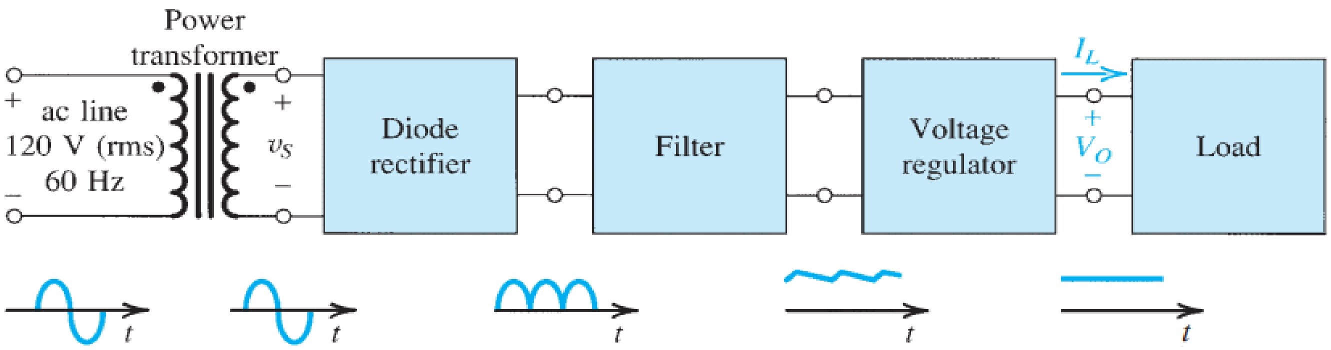


Fig. 1: Block diagram of a dc power supply

- One of the most important applications of diodes is in the design of rectifier circuits.
- A diode rectifier forms an essential building block of the dc power supplies required to power electronic equipment.

Rectifier

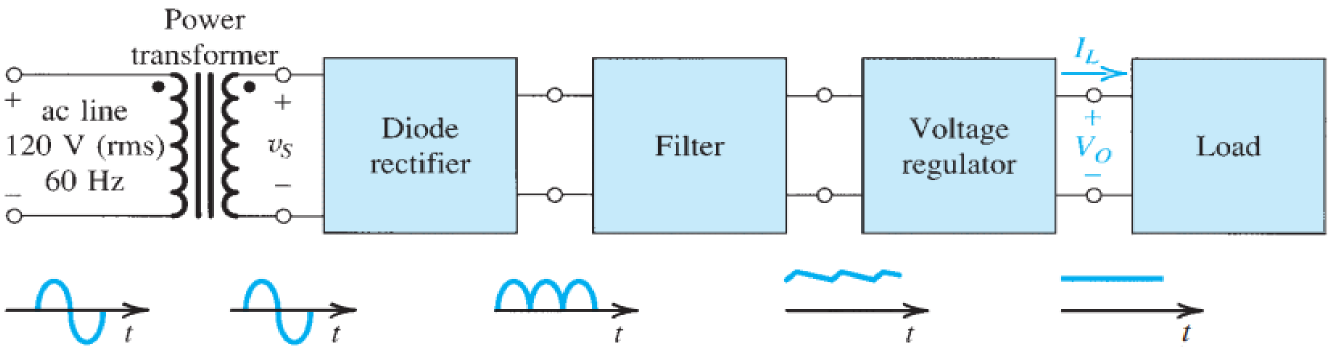


Fig. 1: Block diagram of a dc power supply

- A block diagram of such a power supply is shown in Fig. 1. As indicated, the power supply is fed from the 120-V (rms) 60-Hz ac line, and it delivers a dc voltage V_o (usually in the range of 5 V to 20 V) to an electronic circuit represented by the load block.
- The dc voltage V_o is required to be as constant as possible in spite of variations in the ac line voltage and in the current drawn by the load.

Rectifier Circuits

1. Half-wave Rectifier
2. Full-Wave Rectifier
3. Bridge Rectifier
4. Rectifier with a Filter Capacitor—The Peak Rectifier

Annotation Alert

v_s / v_I = Input AC Voltage

v_o = Output AC Voltage

V_D = Constant DC Voltage Drop of Diode

V_s = Maximum value (+/-) of v_s (DC Value)

V_p = Maximum value of v_o

Half-wave Rectifier

- The half-wave rectifier utilizes alternate half-cycles of the input sinusoid.
- Fig. 2 shows the circuit of a half-wave rectifier.
- Using the more realistic constant-voltage-drop diode

model, we obtain

$$v_O = 0, \qquad v_S < V_D$$

$$v_O = v_S - V_D, \qquad v_S \geq V_t$$

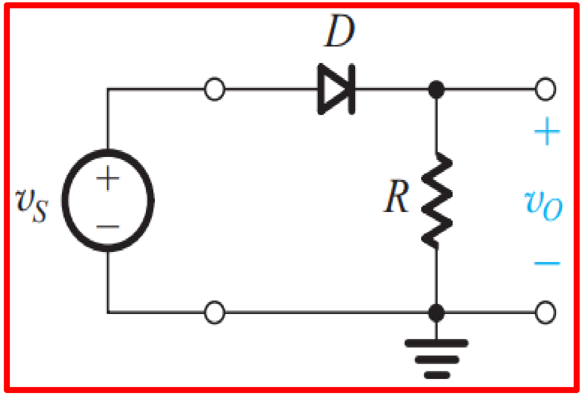


Fig. 2: Half-wave rectifier

BRAC UNIVERSITY

Half-wave Rectifier

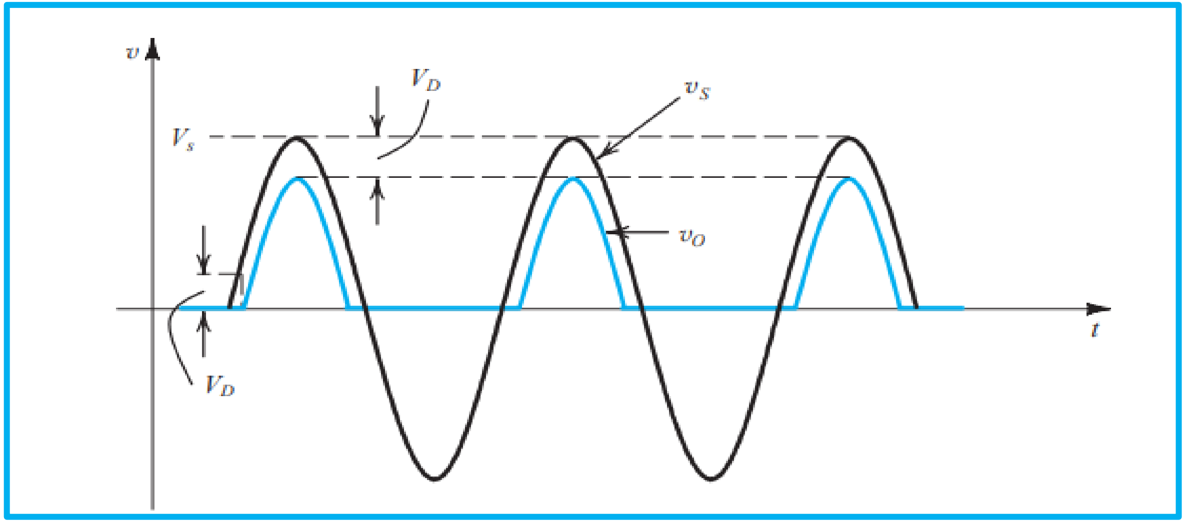
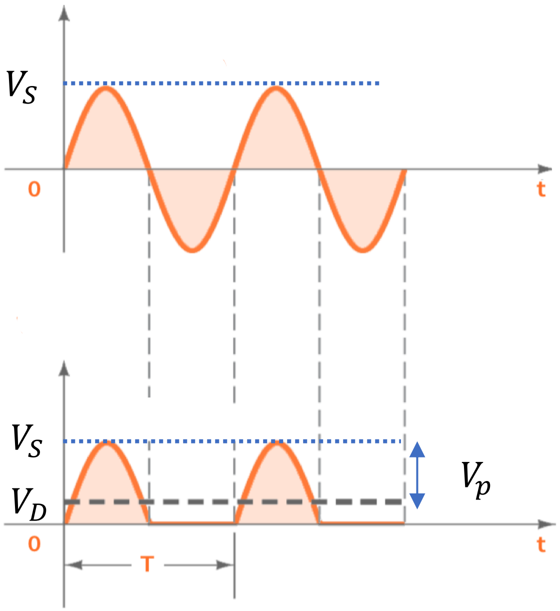


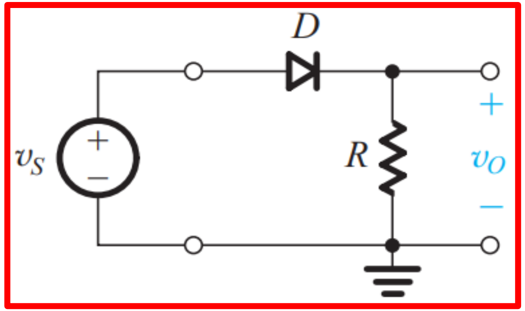
Fig. 3: Input and output waveforms of half-wave rectifier

Half-wave Rectifier

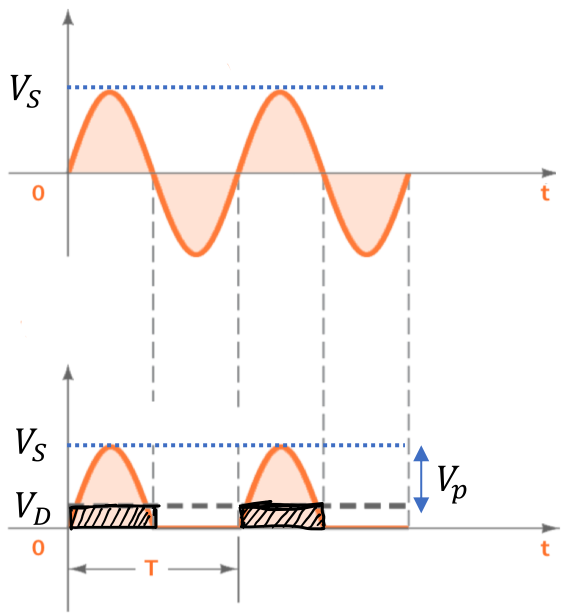


- Suppose, input to the rectifier is $V_s Sin(wt)$ and peak value of output is V_p. Then V_p will be one diode drop below V_s i.e.

$$V_p = V_s - V_D$$

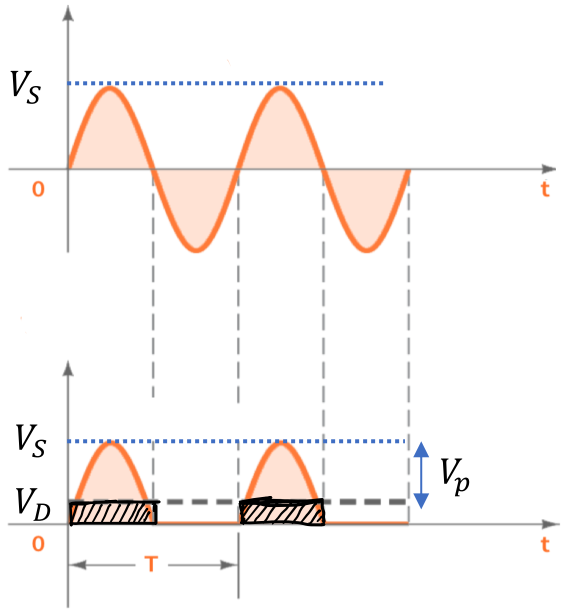


Half-wave Rectifier



- Suppose, Now what is the average value of the output?
- It's not actually straight-forward, since, the output voltage does not start from t = 0. Here, we make an approximation.
- Here, if we subtract the area of the shaded portion, from the area of one of the lobes, we can approximately get the area of the output lobe. Then, averaging over 1 time period will provide the average output voltage.

Half-wave Rectifier



- Area of one lobe of the sine wave,

$$= \int_0^{\frac{T}{2}} V_S Sin(wt)dt$$

$$= \frac{2}{\pi} V_S \left(\frac{T}{2}\right)$$

- Area of the shaded region $= V_D . \left(\frac{T}{2}\right)$
- Approx. area of the output lobe $= \left(\frac{T}{2}\right)[\frac{2}{\pi} V_S - V_D]$
- Average value of voltage (over 1 time period)

$$V_{DC}, or, V_{Avg} = \left(\frac{1}{T}\right) \times \left(\frac{T}{2}\right) \times [\frac{2}{\pi} V_S - V_D] = \frac{1}{\pi} V_S - \frac{1}{2} V_D$$

$$\boxed{V_{DC}, or, V_{Avg} = \frac{1}{\pi} V_S - \frac{1}{2} V_D}$$

Half-wave Rectifier

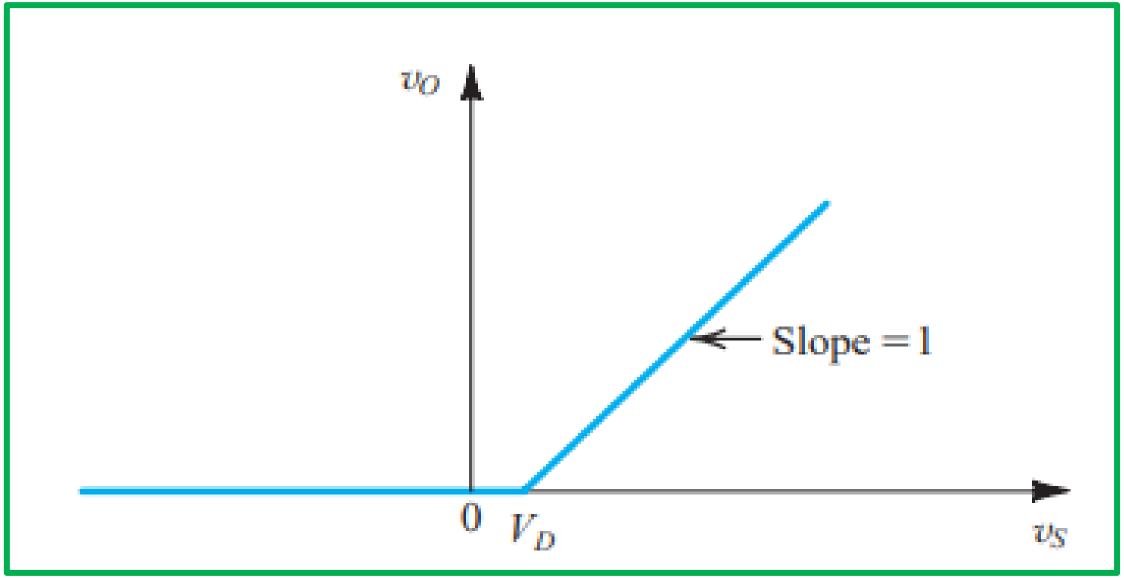


Fig. 4: VTC of half-wave rectifier

Full-wave Bridge Rectifier

- An alternative implementation of the full-wave rectifier is shown in Fig. 8.
- This circuit, known as the bridge rectifier because of the similarity of its configuration to that of the Wheatstone bridge, does not require a center-tapped transformer, a distinct advantage over the full-wave rectifier circuit of Fig. 5.

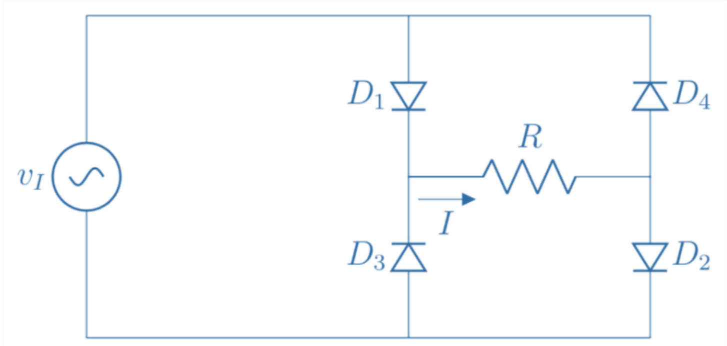


Fig. 5: Full-wave (Bridge) rectifier circuit

Full-wave Bridge Rectifier

- The bridge rectifier, however, requires four diodes as compared to two in the previous circuit. This is not much of a disadvantage

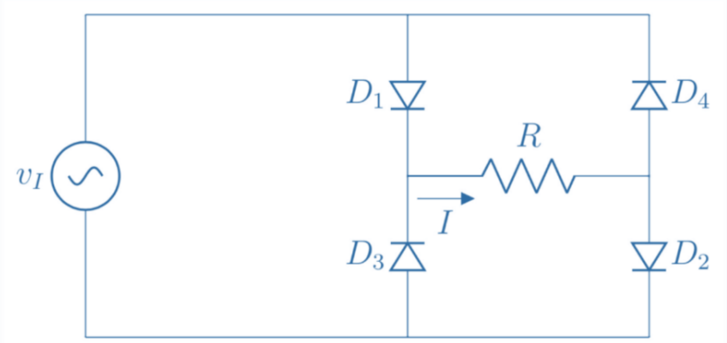


Fig. 5: Full-wave (Bridge) rectifier circuit

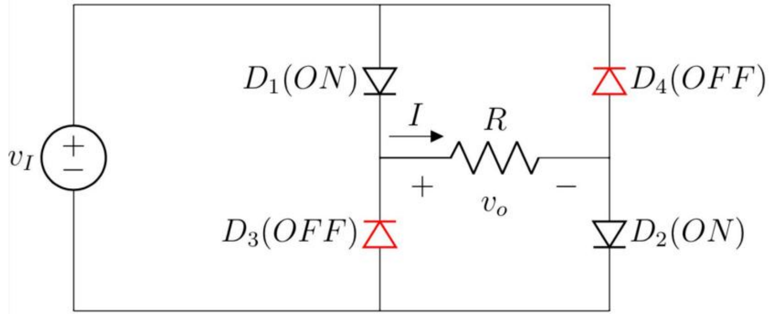
- Because diodes are inexpensive and one can buy a diode bridge in one package.

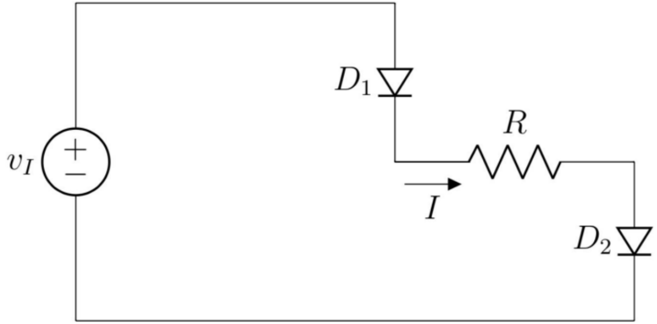
Full-wave Bridge Rectifier

During positive half cycle

- When the input voltage, v_I is positive, and thus current is conducted through diode D_1, resistor R, and diode D_2. Meanwhile, diodes D_3 and D_4 will be reverse biased.
- There are two diodes in series in the conduction path, and thus v_o will be lower than v_I by two diode drops (compared to one drop in the circuit previously discussed).

$$v_o = v_I - 2V_D$$



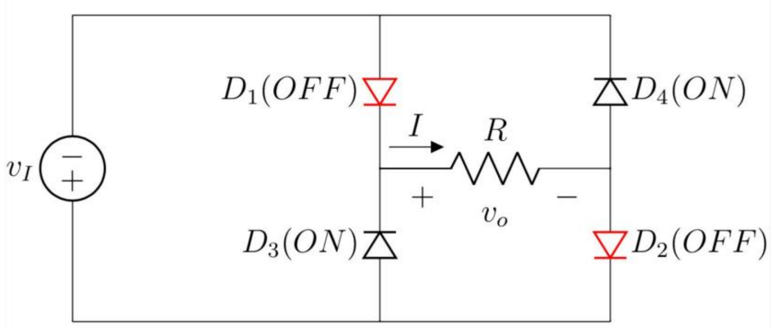


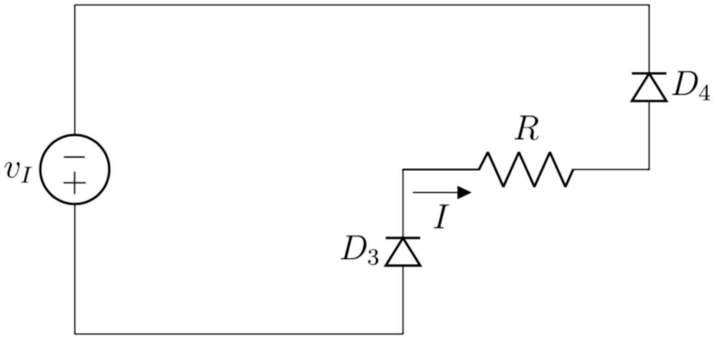
Full-wave Bridge Rectifier

During negative half cycle

- The secondary voltage v_I will be negative, forcing current through D3, R, and D4.
- Meanwhile, diodes D1 and D2 will be reverse biased.
- The important point to note, though, is that during both half-cycles, current flows through R in the same direction (from right to left), and thus v_o will always be positive.

$$v_o = -v_I - 2V_D$$





Full-wave Bridge Rectifier

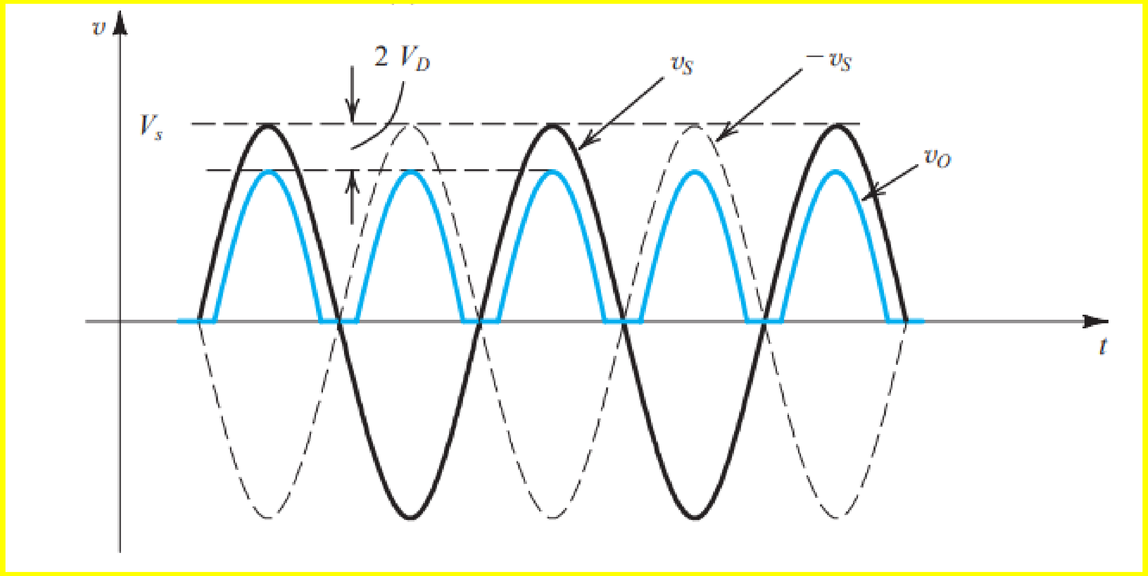
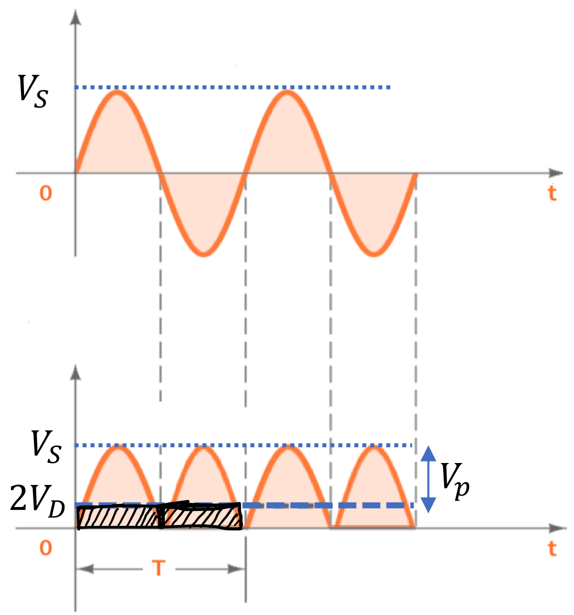


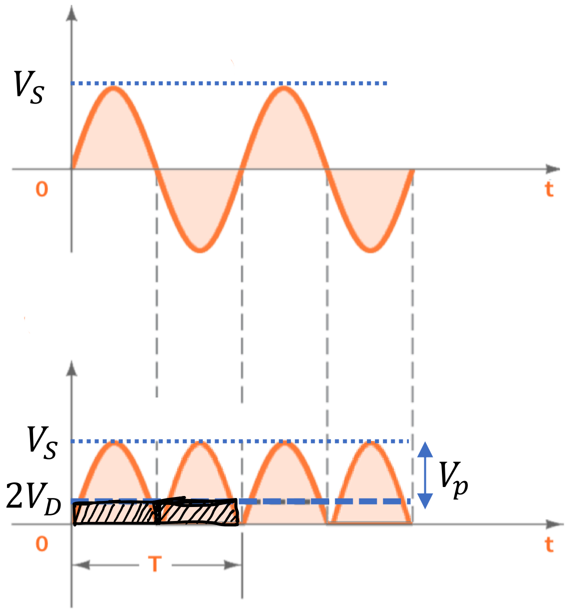
Fig. 6: Input and output waveforms of full-wave (Bridge) rectifier

Full-wave Bridge Rectifier



- Suppose, Now what is the average value of the output?
- It's not actually straight-forward like the previous case, since, the output voltage does not start from t = 0. Here again, we make the same approximation.
- If we subtract the area of the shaded portion, from the area of one of the lobes, we can approximately get the area of the output lobe. Then, averaging over 1 time period will provide the average output voltage.

Full-wave Bridge Rectifier



- Area of two lobes of the sine wave,

$$= \int_0^T V_S Sin(wt)dt$$

$$= \frac{2}{\pi} V_S\ (T)$$

- Area of the shaded region = $2V_D.T$
- Approx. area of the output lobes = $T[\frac{2}{\pi} V_S\ - 2V_D]$
- Average value of voltage (over 1 time period)

$$V_{DC}, or, V_{Avg} = \left(\frac{1}{T}\right) \times (T) \times [\frac{2}{\pi} V_S\ - 2V_D] = \frac{2}{\pi} V_S - 2V_D$$

$$\boxed{V_{DC}, or, V_{Avg} = \frac{2}{\pi} V_S\ - 2V_D}$$

Full-wave Bridge Rectifier

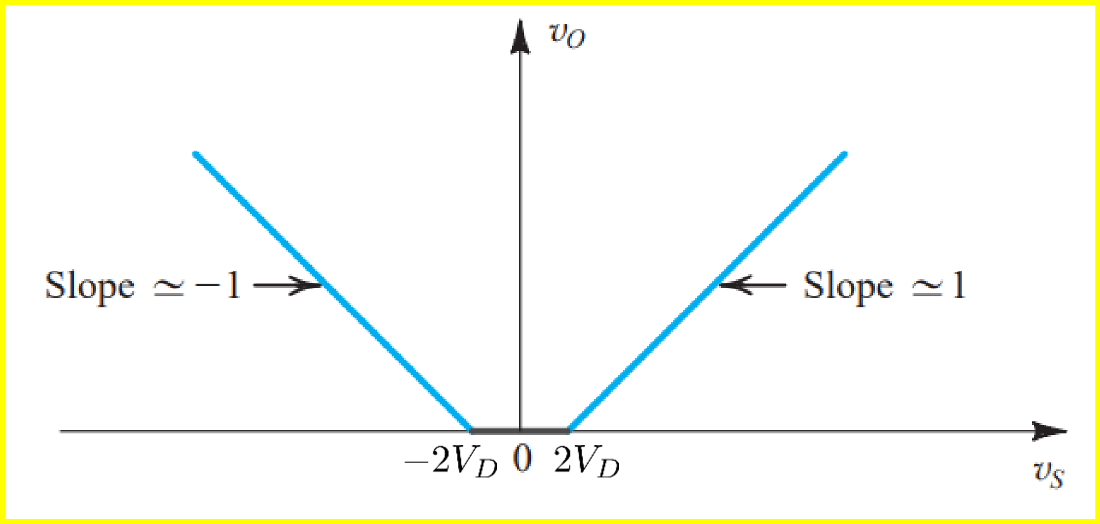
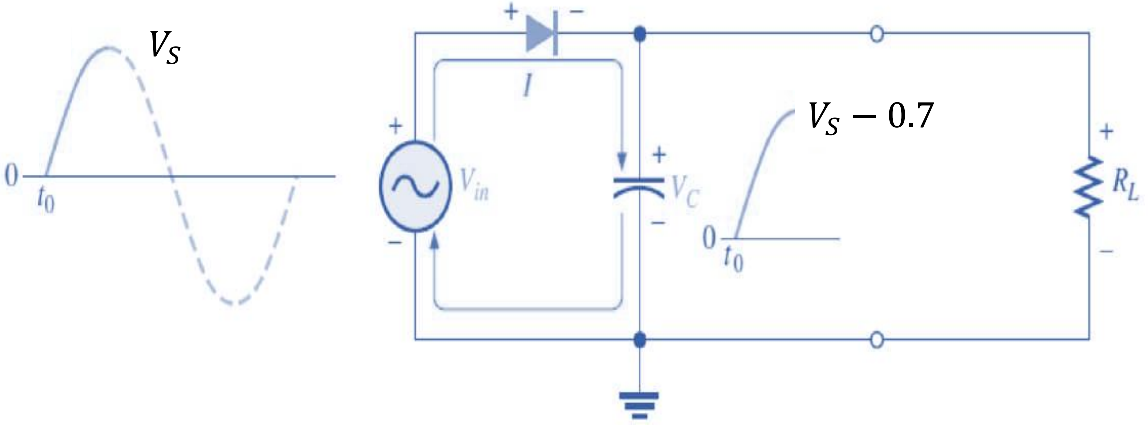


Fig. 10: VTC of full-wave (Bridge) rectifier

HW Rectifier with Filter Capacitor

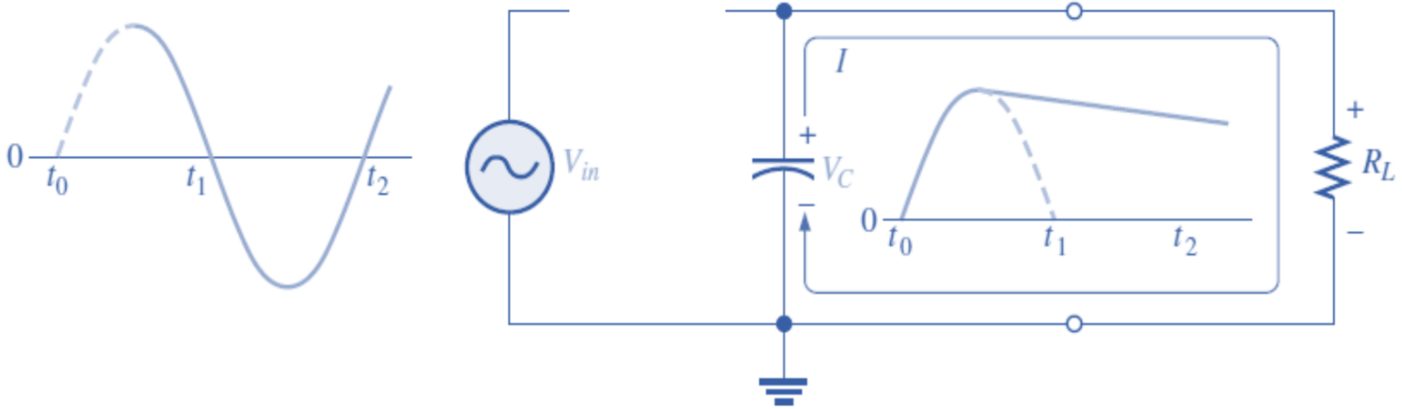
A half-wave rectifier with a capacitor-input filter is shown in part (a). During the positive first quarter-cycle of the input, the diode is forward-biased, allowing the capacitor to charge to within 0.7 V of the input peak, as illustrated in part (a).



(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on

HW Rectifier with Filter Capacitor

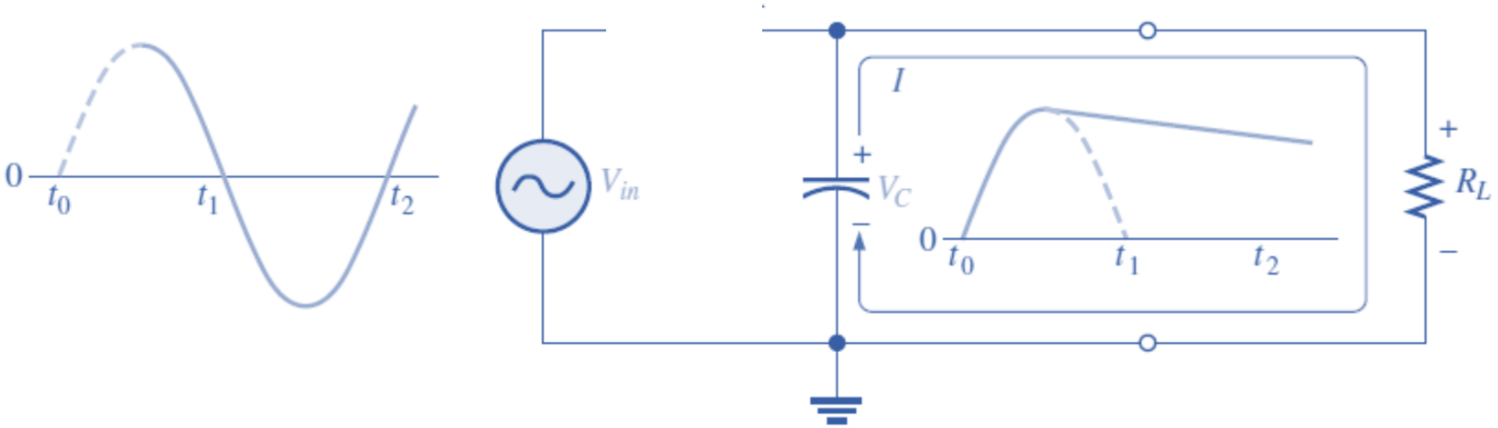
When the input begins to decrease below its peak, as shown in part (b), the capacitor retains its charge and the diode becomes reverse-biased because the cathode is more positive than the anode.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.

HW Rectifier with Filter Capacitor

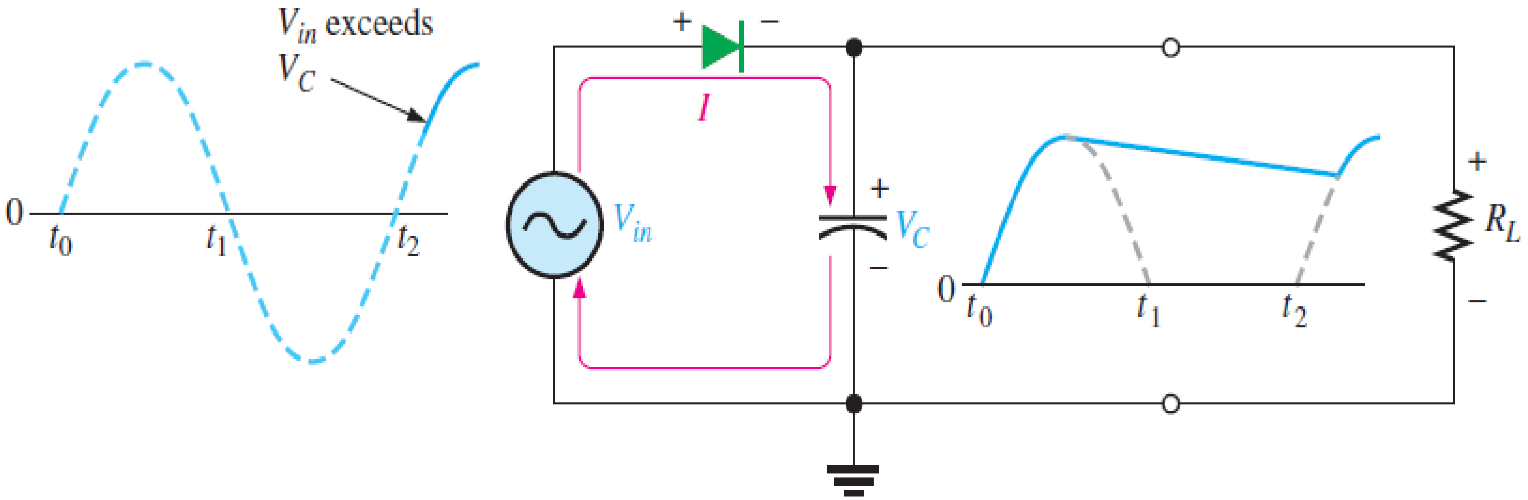
During the remaining part of the cycle, the capacitor can discharge only through the load resistance at a rate determined by the $R_L C$ time constant, which is normally long compared to the period of the input.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.

HW Rectifier with Filter Capacitor

(a), (b), (c) depict the operation of a half-wave rectifier with a capacitor-input filter. The current indicates charging or discharging of the capacitor.

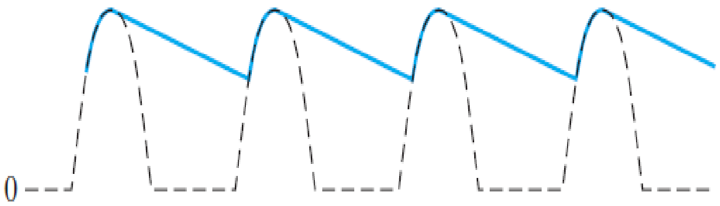


(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid dark blue curve.

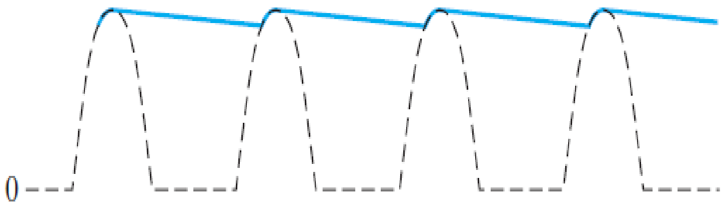
HW Rectifier with Filter Capacitor

Ripple Voltage

As you have seen, the capacitor quickly charges at the beginning of a cycle and slowly discharges through R_L after the positive peak of the input voltage (when the diode is reverse-biased). The variation in the capacitor voltage due to the charging and discharging is called the **ripple voltage**. Generally, ripple is undesirable; thus, the smaller the ripple, the better the filtering action, as illustrated in below figures.



(a) Larger ripple (blue) means less effective filtering.



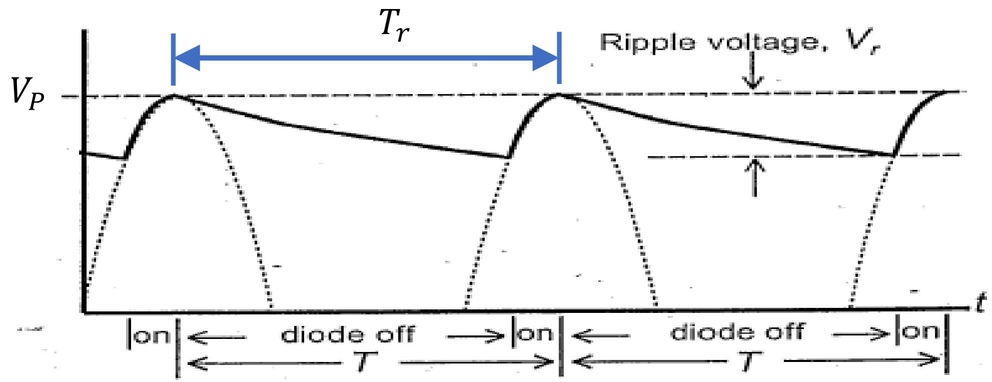
(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

HW Rectifier with Filter Capacitor

Ripple Voltage

- The peak-to-peak ripple voltage is, $V_r\ (p-p) = \frac{V_P}{f_r RC}$ (Please refer to here for details derivation)

Where, f_r is ripple frequency and $f_r = \frac{1}{T_r}$. Note that, $f_r = f_s$ which is the input signal frequency

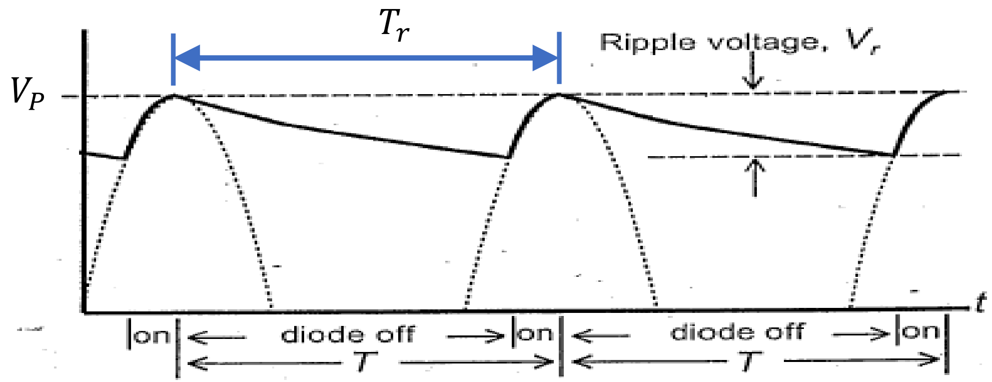


HW Rectifier with Filter Capacitor

Ripple Voltage

- We can write the output voltage $V_{out}(t) = V_{out}(AC) + V_{out}(DC)$
- $V_{out}(DC)$ or V_{DC} will go through the middle of the ripple. So, we can write-

$$V_{out}(DC) = V_P - \frac{V_{r(p-p)}}{2}$$

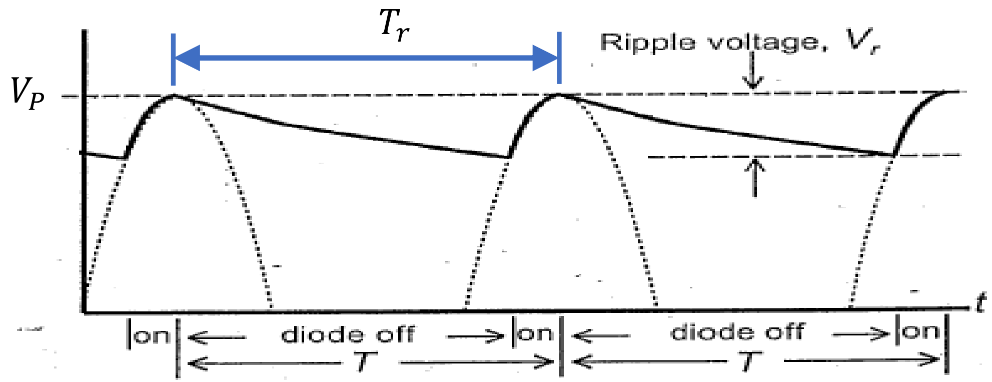


HW Rectifier with Filter Capacitor

Ripple Voltage

- The AC value/ R.M.S value of the ripple voltage can be written as

$$V_r(AC) \text{ or } V_{AC} = \frac{V_r(p-p)}{2\sqrt{3}} = \frac{V_P}{2\sqrt{3} f_r RC}$$



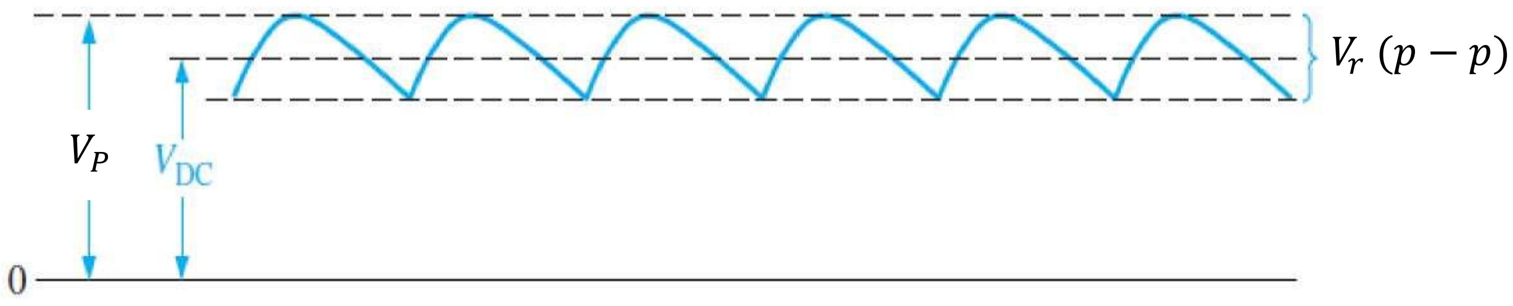
HW Rectifier with Filter Capacitor

Ripple Factor

The **ripple factor (r)** is an indication of the effectiveness of the filter and is defined as

$$r = \frac{V_{AC}}{V_{DC}}$$

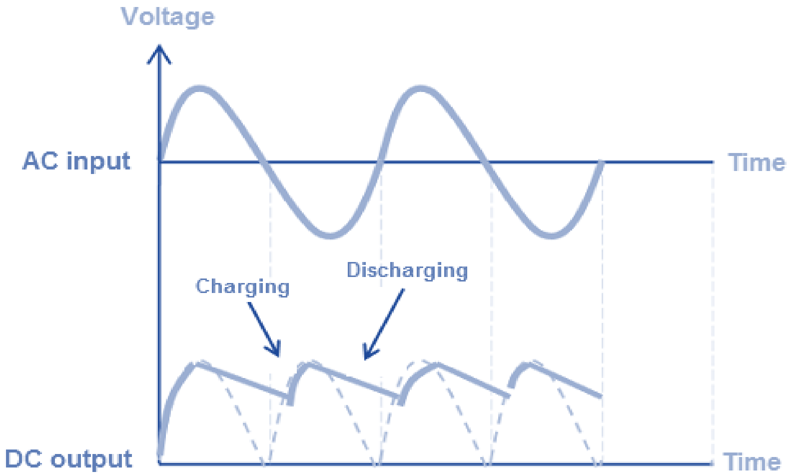
where V_{AC} is the RMS ripple voltage and V_{DC} is the dc (average) value of the filter's output voltage, as illustrated in below figure. The lower the ripple factor, the better the filter. The ripple factor can be lowered by increasing the value of the filter capacitor or increasing the load resistance.



FW Rectifier with Filter Capacitor

The ripple analysis of a full-wave (FW) rectifier is essentially the same as that of a half-wave (HW) rectifier, with one key difference. As illustrated in the following figure, the ripple period for the FW rectifier is half that of the input signal. This occurs because the capacitor begins to charge when the input reaches its negative peak. So.

$$f_r = 2f_s \quad \text{and} \quad T_r = \frac{T_s}{2}$$



To Sum Up

Param.	HW	FW
V_P	$V_M - V_D$	$V_M - 2V_D$
f_r	f_s	$2f_s$
V_{DC} (Without Cap.)	$\frac{1}{\pi} V_S - \frac{1}{2} V_D$	$\frac{2}{\pi} V_S - 2V_D$

With Capacitor
$V_r\ (p-p) = \frac{V_P}{f_r RC}$
$V_r(AC)$ or $V_{AC} = \frac{V_r(p-p)}{2\sqrt{3}}$
$V_r(AC)$ or $V_{AC} = \frac{V_P}{2\sqrt{3} f_r RC}$
$V_{out}(DC) = V_P - \frac{V_r(p-p)}{2}$

Practice Problem 1

The input voltage of a half-wave rectifier is a **Sine** wave with amplitude $V_M = 10$ V and 40 Hz frequency. The output load resistance is $R = 50\ \text{k}\Omega$. A silicon diode is used in this circuit whose forward voltage drop is $V_{D_0} = 0.6$ V.

(a) [**4 marks**] Briefly **explain** the purpose of a rectifier. **Draw** the input and output waveforms of the mentioned rectifier with proper labeling. **Draw** the Voltage Transfer Characteristic (VTC) curve of the rectifier.

(b) [**1 mark**] **Calculate** the DC value of the output voltage, V_{DC}.

(c) [**3 marks**] After connecting a capacitor parallel to the load, V_{DC} changes to $V_{\text{DC(Cap)}}$ and the new output voltage can be expressed as $V_{\text{out}} = V_{\text{DC(Cap)}} \pm 0.5$ V. **Calculate** the peak-to-peak ripple voltage $V_{r(p-p)}$, and **determine** the ripple frequency, f_{r}. Now, **calculate** the value of the capacitor, C.

(d) [**1 mark**] **Calculate** the DC value of the output voltage after connecting the capacitor, $V_{\text{DC(Cap)}}$.

(e) [**1 mark**] **Compare** $V_{\text{DC(Cap)}}$ with V_{DC} determined in part-(b) and briefly **explain** their difference.

Practice Problem 2

A voltage waveform $v_i = 5\sin(200\pi t)$ V is fed into a Full-wave rectifier with a load resistor, $R = 5$ kΩ. Silicon diodes are used in this circuit where, $V_{D_0} = 0.6$ V.

(a) **Draw** the rectifier circuit. **Label** the input and output voltages properly. Briefly **explain** the application of the circuit. [1+1+1]

(b) **Calculate** the DC value of the output voltage, V_{dc} and the output frequency, f_o. [1+1]

(c) **Draw** the Voltage Transfer Characteristics (VTC) of the Full-wave rectifier and **label** it properly. [2]

(d) Now, you have to connect a capacitor in parallel with the load resistor. You have two capacitors of 5 μF and 1 μF at your disposal. Which capacitor will you use? **Explain** briefly with necessary calculations. [3]

(e) [**Bonus**] A different input waveform is fed into the Full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated from (d) with the 5 μF capacitor. The new output frequency is 300 Hz. **Determine** the equation of the input waveform. [2]

Practice Problem 3

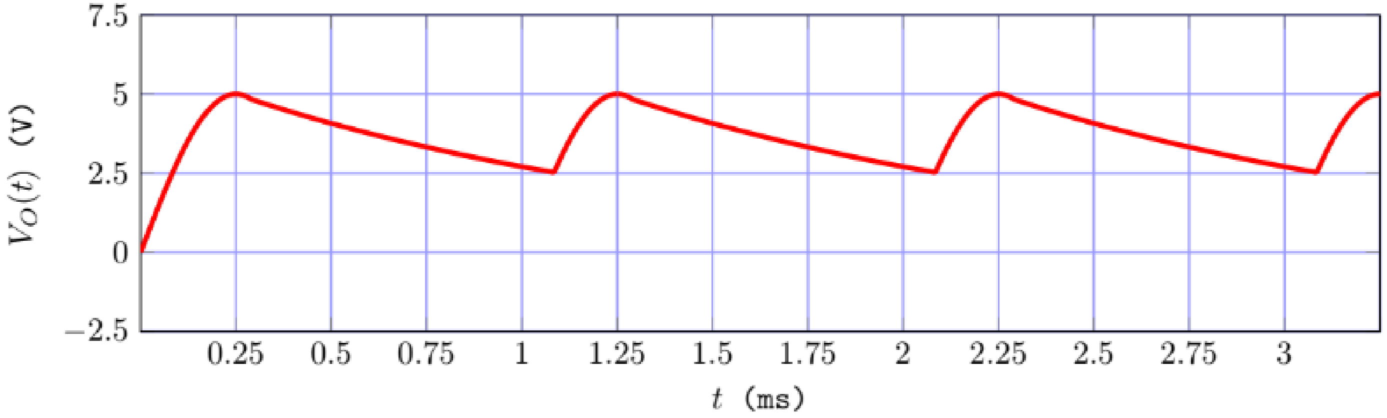


Figure-3

(c) **[2 marks]** In *Figure-3*, you are given the output voltage waveform of an unknown rectifier circuit with an output load resistance, $R = 5k\Omega$, input frequency, $f_{in} = 0.5kHz$, and $V_{D0} = 0.55$V. **Analyze** the waveform in *Figure-3*, and **determine** the output voltage frequency, f_{out} and **draw** the rectifier circuit with proper labels.

(d) **[2 marks]** **Analyze** the circuit in *Figure-4*, and **draw** the output voltage waveform for $V_{in} = 6sin(100\pi t)\ V$.

Practice Problem 4

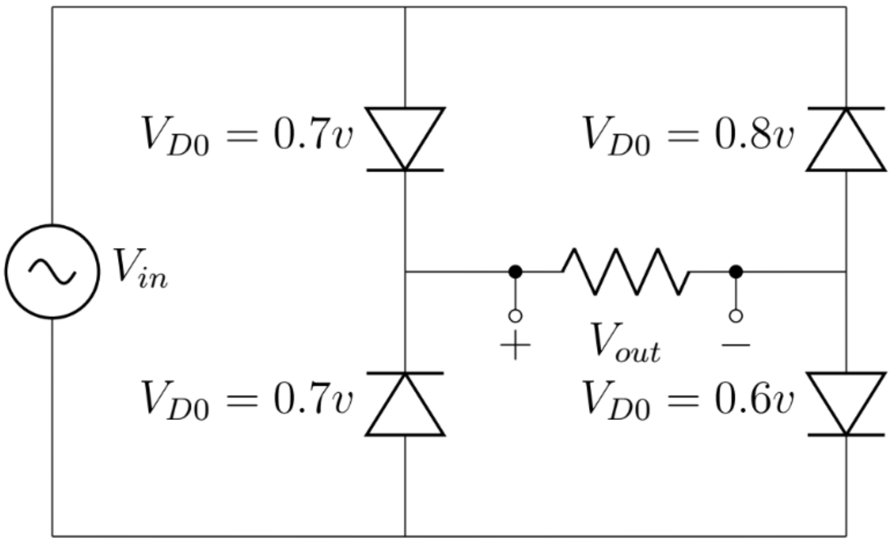


Figure-4

Analyze the circuit in *Figure-4*, and **draw** the output voltage waveform for $V_{in} = 6sin(100\pi t)\ V$.

Practice Problems More

For more problems, please refer to the following link: Practice Problems

Slide Acknowledgement

1. Microelectronic Circuits by Adel S. Sedra & Kenneth C. Smith: Chapter 4. 7th edition. Published by Oxford University Press.

Thank you for your attention